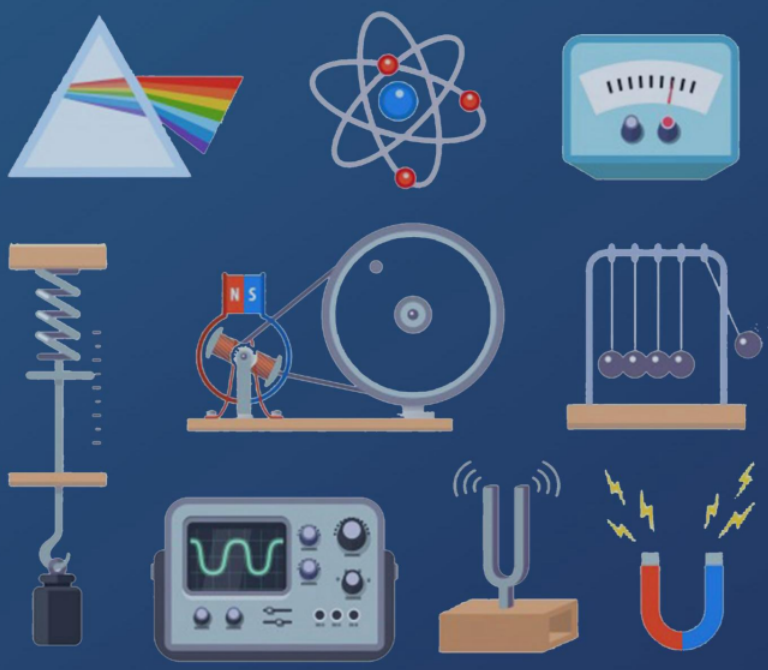
PHYSICS

Current Electricity



SHEET

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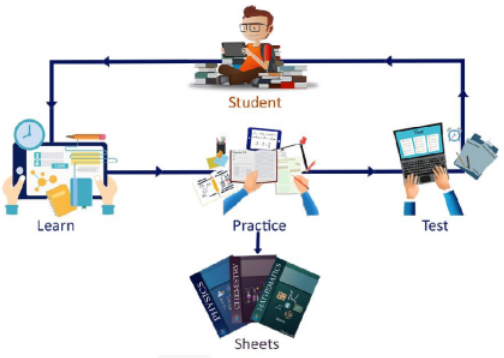
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Electric Current	1	12		
2	Drift Velocity	2-8			
3	Ohm's Law	9		1	
4	Resistance and Resistors	10-14	13	2	
5	Battery And Emf	15	1,8,14		
6	Kirchhoff's Laws		9		
7	Series And Parallel Combination	16-19	15	3	
8	Circuit Solving	20,21	2,4-6,10	4	
9	Wheatstone Bridge	22		5	1
10	Series And Parallel Combination Of Batteries	23-26		6	
11	Power Delivered and Heat Dissipated	27-34		7-13	2-5
12	Using Ammeter and voltmeter	35-38	11	14-16	6
13	Constructing Ammeter and Voltmeter		16	17	7
14	Meter Bridge			18-20	8
15	Potentiometer	39,40	7	21	9-13,22
16	Post Office Box	41			
17	After full Chapter		3	22	14-21,23

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Electric Current

Q1. In the Bohr's model of hydrogen atom, the electron moves around the nucleus in a circular orbit of radius 5×10^{-11} m. Its time period is 1.5×10^{-16} s. The current associated with the electron motion is :-

(A) zero (B) 1.6×10^{-19} A
(C) 0.17 A (D) 1.07×10^{-3} A

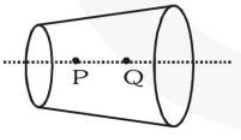
Drift Velocity

Q2. The number of free electrons per 10 mm of an ordinary copper wire is about 2×10^{21}. The average drift speed of the electrons is 0.25 mm/sec The current flowing is :-

(A) 0.8A (B) 8A
(C) 80A (D) 5A

Q3. A wire has a non–uniform cross–section as shown in figure. A steady current flows through it. The drift speed of electrons at points P and Q is v_P and v_Q, then

(A) $v_P = v_Q$
(B) $v_P < v_Q$
(C) $v_P > v_Q$
(D) data insufficient



Q4. There are 8.4×10^{22} free electrons per cm^3 in copper. The current in the wire is 0.21 A, then the drift velocity of electrons in a copper wire of $1mm^2$ cross-section will be $\left(e = 1.6 \times 10^{-19} C\right)$:-

(A) 2.12×10^{-5} m/s (B) 0.78×10^{-5} m/s
(C) 1.56×10^{-5} m/s (D) none of these

Q5. **Statement-1 :** The electric bulb glows immediately when switch is on.

Statement-2 : The drift velocity of electrons in a metallic wire is very high.

(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.
(B) Statement–1 is True, Statement–2 is True; Statement–2 is NOT a correct explanation for Statement–1.
(C) Statement–1 is True, Statement–2 is False.
(D) Statement–1 is False, Statement–2 is True.

Q6. An insulating pipe of cross-section area 'A' contains an electrolyte which has two types of ions → their charges being –e and +2e. A potential difference applied between the ends of the pipe result in the drifting of the two types of ions, having drift speed = v (–ve ion) and v/4 (+ve ion). Both ions have the same number per unit volume = n. The current flowing through the pipe is

(A) $\frac{1}{2}$ nev A (B) $\frac{1}{4}$ nev A
(C) $\frac{5}{2}$ nev A (D) $\frac{3}{2}$ nev A

Q7. A uniform copper wire carries a current i amperes and has p carriers per $metre^3$. The length of the wire is ℓ metres and its cross-section area is s $metre^2$. If the charge on a carrier is q coulombs, the drift velocity in ms^{-1} is given by

(A) $i/\ell sq$ (B) i/psq
(C) psq/i (D) $i/ps\ell q$

Q8. A current I flows through a uniform wire of diameter d when the mean electron drift velocity is v. The same current will flow through a wire of diameter d/2 made of the same material if the mean drift velocity of the electron is :

(A) v/4 (B) v/2
(C) 2v (D) 4v

Ohm's Law

Q9. **Statement–1 :** A metal has resistance and gets often heated by flow of current.

Statement–2 : When free electrons drift through a metal they makes occasional collisions with the lattice. These collisions are inelastic and transfer energy to the lattice as internal energy.

(A) Statement–1 is True, Statement–2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement–1 is True, Statement–2 is True; Statement-2 is NOT a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

Resistance and Resistors

Q10. Three copper wires of lengths and cross-sectional areas are (ℓ, A); $\left(2\ell, \frac{A}{2}\right)$ and $\left(\frac{\ell}{2}, 2A\right)$. Resistance will be minimum in:-

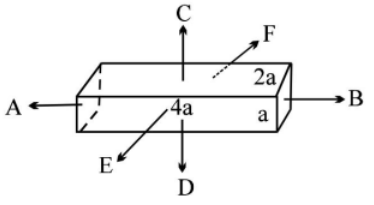
(A) wire of cross-sectional area $\frac{A}{2}$

(B) wire of cross-sectional area A

(C) wire of cross-sectional area 2A

(D) same is all the three cases

Q11. A conductor with rectangular cross section has dimensions (a×2a×4a) as shown in figure. Resistance across AB is x, across CD is y and across EF is z. Then -



(A) x = y = z (B) x > y > z

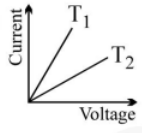
(C) y > x > z (D) x > z > y

Q12. When a wire of uniform cross-section a, length ℓ and resistance R is bent into a complete circle, resistance between any two of diametrically opposite points will be :–

(A) $\frac{R}{2}$ (B) $\frac{R}{4}$

(C) $\frac{R}{8}$ (D) 4R

Q13. The current in a metallic conductor is plotted against voltage at two different temperatures T_1 and T_2. Which is correct



(A) $T_1 > T_2$ (B) $T_1 < T_2$

(C) $T_1 = T_2$ (D) none

Q14. Two resistance R_1 and R_2 are made of different materials. The temperature coefficient of the material of R_1 is α and of the material of R_2 is $-\beta$. The resistance of the series combination of R_1 and R_2 will not change with temp., then ratio of resistance of two wire at 0°C will be :

(A) $\frac{\alpha}{\beta}$ (B) $\frac{\alpha+\beta}{\alpha-\beta}$

(C) $\frac{\alpha^2+\beta^2}{\alpha\beta}$ (D) $\frac{\beta}{\alpha}$

Battery And Emf

Q15. A battery has e.m.f. 4V and internal resistance 'r'. When this battery is connected to an external resistance of 2 ohms, a current of 1 amp. flows in the circuit. How much current will flow if the terminals of the battery are connected directly :–

(A) 1 A (B) 2 A

(C) 4 A (D) infinite

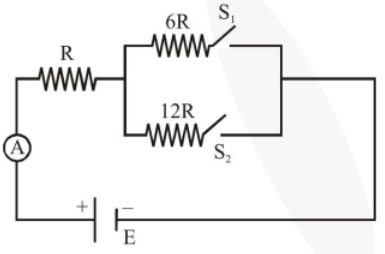
Series And Parallel Combination

Q16. The equivalent resistance of a group of resistances is R. If another resistance is connected in parallel to the group, its new equivalent becomes R_1 and if it is connected in series to the group, it new equivalent becomes R_2 we have :

(A) $R_1 > R$ or $R_2 > R$ (B) $R_1 < R$ or $R_2 > R$

(C) $R_1 > R$ or $R_2 < R$ (D) $R_1 < R$ or $R_2 < R$

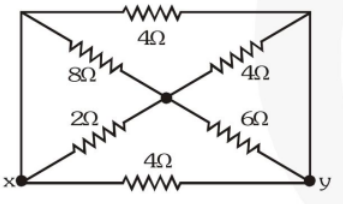
Q17. When only S_1 is closed then reading of ammeter is I_1, when only S_2 is closed then reading I_2, when both S_1 & S_2 are closed simultaneously then reading is I_3 then :-



(A) $I_1 > I_2 > I_3$ (B) $I_2 > I_3 > I_1$

(C) $I_3 > I_2 > I_1$ (D) $I_3 > I_1 > I_2$

Q18. The total resistance between x and y in ohm is :–



(A) 1 Ω (B) 4 Ω

(C) (4/3) Ω (D) (2/3) Ω

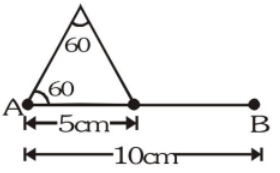
Q19. A wire has resistance of 24 Ω is bent in the following shape. The effective resistance between A and B is-

(A) 24 Ω

(B) 10 Ω

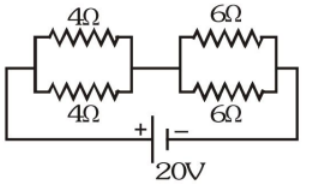
(C) $\frac{16}{3}\Omega$

(D) None of these



Circuit Solving

Q20. Four resistances are connected in a circuit as shown in the following diagram. The value of the current in ampere in 4Ω and 6Ω resistance are :-



(A) 2 and 4 (B) 1 and 2

(C) 1and 1 (D) 2 and 2

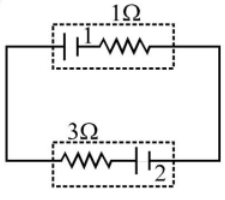
Q21. In the figure shown, battery 1 has emf = 6 V and internal resistance = 1 Ω. Battery 2 has emf = 2V and internal resistance = 3 Ω. The wires have negligible resistance. What is the potential difference across the terminals of battery 2 ?

(A) 4 V

(B) 1.5 V

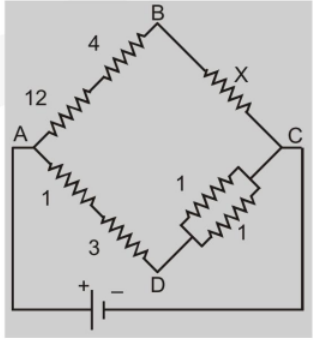
(C) 5 V

(D) 0.5 V



Wheatstone Bridge

Q22. In the arrangement of resistances shown in the circuit, the potential difference between points B and D will be zero, when the unknown resistance X is :-

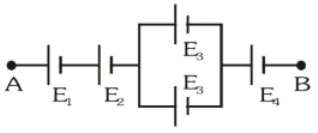


(A) 4 Ω (B) 3 Ω

(C) 2 Ω (D) 1 Ω

Series And Parallel Combination Of Batteries

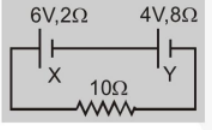
Q23. In the following circuit the resultant e.m.f. between AB is :–



(A) $E_1+E_2+E_3+E_4$ (B) $E_1+E_2+2E_3+E_4$

(C) $E_1+E_2+\frac{E_3}{2}+E_4$ (D) $E_1+E_2+\frac{E_3}{4}+E_4$

Q24. Two cells X and Y are connected to a resistance of 10Ω as shown in the figure. The terminal voltage of cell Y is :-



(A) zero (B) 2V

(C) 4V (D) 10V

Q25. 100 cells, each of e.m.f. 5 V and internal resitance 1 Ω, are to be arranged so as to produce maximum current in a 25 Ω resistance. Each row is to contain equal number of cells. The number of rows should be :-

(A) 2 (B) 4

(C) 5 (D) 10

Q26. Two nonideal batteries are connected in parallel consider the following statements:-

(A) The equivalent emf is smaller than either of the two emfs

(B) The equivalent internal resistance is smaller than either of the two internal resistances

(A) both A and B are correct

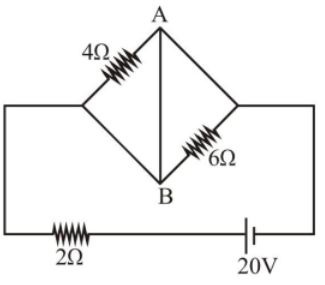
(B) A is correct but B is wrong

(C) B is correct but A is wrong

(D) both A and B are wrong

Power Delivered and Heat Dissipated

Q27. In the circuit shown in figure. Which of the following is the incorrect.



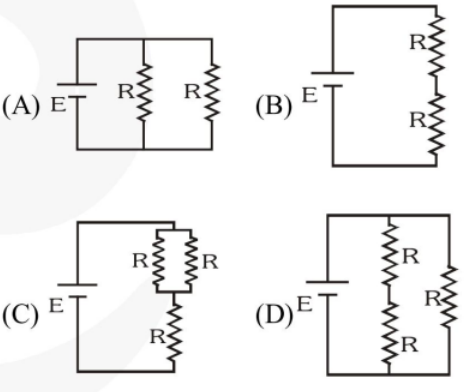
(A) Power supplied by the battery is 200W

(B) Current flowing in the circuit is 5A

(C) Potential difference across the 4Ω resistance is equal to the potential difference across the 6Ω resistance.

(D) Current in wire AB is not zero.

Q28. Consider four circuits shown in the figure below. In which circuit power dissipated is greater (Neglect the internal resistance of the power supply) :–



Q29. When a constant voltage is applied between the ends of a metal wire. Heat produced in this wire will be doubled if:

(A) both the length and the radius of the wire are halved.

(B) both the length and the radius of the wire are doubled.

(C) the radius of the wire is doubled.

(D) the length of the wire is doubled.

Q30. A lamp of 100 W and a heater of 1 kW are in simultaneous use for 10 hours. The units of electricity consumed according to the meter in the house is :

(A) 10 (B) 9 (C) 11 (D) 1

Q31. A 60 watt bulb operates on 220 V supply. The current flowing through the bulb is :-

(A) $\frac{11}{3}$A (B) $\frac{3}{11}$A

(C) 8 A. (D) 6 A.

Q32. An electric bulb is designed to draw power P_0 at voltage V_0. If the voltage is V, it draws a power P, then –

(A) $P = \left(\frac{V}{V_0}\right)P_0$ (B) $P = \left(\frac{V_0}{V}\right)P_0$

(C) $P = \left(\frac{V_0}{V}\right)^2 P_0$ (D) $P = \left(\frac{V}{V_0}\right)^2 P_0$

Q33. 'n' identical bulbs, each designed to draw a power P from a certain voltage supply are joined in series across the supply. The total power which they will draw is :–

(A) P (B) nP (C) $\frac{P}{n^2}$ (D) $\frac{P}{n}$

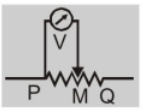
Q34. A battery of 10 V and internal resistance 0.5 Ω is conneted across a variable resistance R. The value of R for which the power delivered in its maximum state, is equal to:–

(A) 0.5 Ω (B) 1 Ω

(C) 1.5 Ω (D) 2 Ω

Using Ammeter and voltmeter

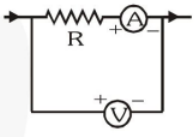
Q35. P, Q is a uniform wire of resistance 2000 Ω and M the mid point of PQ. A voltmeter of resistance 1000 Ω is connected between P and M. The reading of the voltmeter, when, the potential difference applied between PQ is 150 volt will be.



(A) 150 volt (B) 100 volt

(C) 75 volt (D) 50 volt

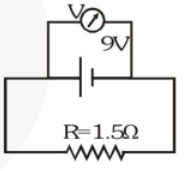
Q36. In the following circuit, the resistance of a voltmeter is 10,000 Ω and that of an ammeter is 20 Ω. If the reading of an ammeter is 0.1 amp. and that of voltmeter is 12 volt, then the value of R is :–



(A) 122 Ω (B) 100 Ω

(C) 118 Ω (D) 116 Ω

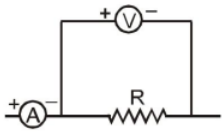
Q37. When a voltmeter is conected across the terminals of a cell, it measure 9 Volt. If a resistance of 1.5 Ω is connected across the terminals of a cell as shown in figure. Then current flowing through this resistance is :–



(A) 1.3 A (B) 6 A

(C) 9 A (D) 12 A

Q38. A student connects a non ideal voltmeter, ammeter and resistance according to the circuit given. If the voltmeter reading is 20 V and ammeter reading is 4 A, then the resistance will be :-



(A) equal to 5 Ω (B) more than 5 Ω

(C) less than 5 Ω (D) equal to 2 Ω

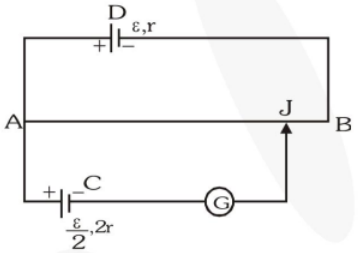
Potentiometer

Q39. A potentiometer wire has length 10 m and resistance 10Ω. It is connected to a battery of EMF 11 volt and internal resistance 1 Ω, then the potential gradient in the wire is

(A) 10 V/m (B) 1 V/m

(C) 0.1 V/m (D) none

Q40. In the figure, the potentiometer wire AB of length L and resistance 9r is joined to the cell D of emf ε and internal resistance r. The cell C's emf is $\frac{\varepsilon}{2}$ and its internal resistance is 2r. The galvanometer G will show no deflection when the length AJ is -

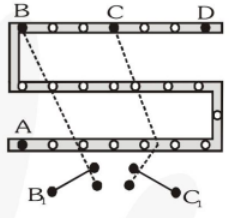


(A) $\frac{4L}{9}$ (B) $\frac{5L}{9}$

(C) $\frac{7L}{18}$ (D) $\frac{11L}{18}$

Post Office Box

Q41. For the post office box arrangement to determine the value of unknown resistance, the unknown resistance should be connected between.



(A) B and C (B) C and D

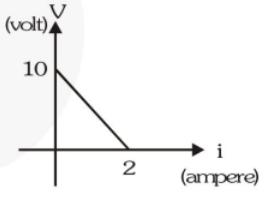
(C) A and D (D) B_1 and C

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A battery of emf E and internal resistance r is connected across a resistance R. Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current (i) passing through the resistance and potential difference (V) across it. Select the correct alternative(s).



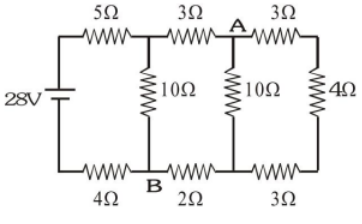
(A) Internal resistance of battery is 5Ω

(B) Emf of the battery is 20V

(C) Maximum current which can be taken from the battery is 4A

(D) V–i graph can never be a straight line as shown in figure

Q2. Consider the circuit shown in the figure :



(A) The current in the 5Ω resistor is 2A
(B) The current in the 5Ω resistor is 1A
(C) The potential difference $V_A - V_B$ is 7V
(D) The potential difference $V_A - V_B$ is 5V

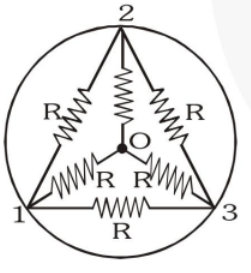
Q3. Two identical fuses are rated at 10A. If they are joined
(A) in parallel, the combination acts as a fuse of rating 20A
(B) in parallel, the combination acts as a fuse of rating 5A
(C) in series, the combination acts as a fuse of rating 10A
(D) in series, the combination acts as a fuse of rating 20A

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 4 to 6

Inside a super conducting ring six identical resistors each of resistance R are connected as shown in figure:



Q4. The equivalent resistance(s).
(A) between 1 & 3 is zero
(B) between 1 & 3 is R/2
(C) between 1 & 2, 2 & 3, 3 & 1 are all equal
(D) None of these

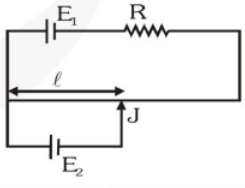
Q5. The equivalent resistance (s).
(A) between 0 & 1 is R
(B) between 0 & 1 is R/3
(C) between 0 & 1 is 0
(D) between 0 & 1, 0 & 2 and 0 & 3 are all equal

Q6. Imaging a battery of emf E between the point 0 and 1, with its positive terminal connected with O.
(A) The current entering at O is equally divided into three resistances.
(B) The current in the other three resistances R_{12}, R_{13}, R_{23} is zero.
(C) The resistances R_{02} and R_{03} have equal magnitudes of current while the resistance R_{01} have different current.
(D) Potential $V_2 = V_3 > V_1$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q7. In the potentiometer arrangement shown in figure, null point is obtained at length ℓ.

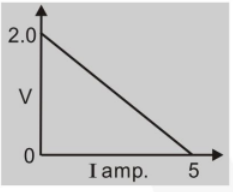


Column I	Column II
(A) If E_1 is increased	(P) ℓ should increase
(B) If R is increased	(Q) ℓ should decrease
(C) If E_2 is increased	(R) ℓ should remain the same to again get the null point

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

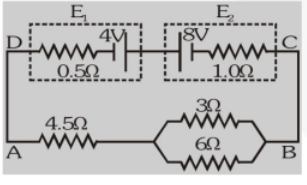
Q8. For a cell, the graph between the potential difference (V) across the terminals of the cells and the current I drawn from the cell is as shown in figure. Calculate the e.m.f. and the internal resistance of the cell.



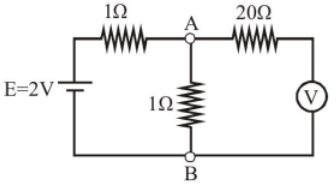
Q9. What is the value of current I in the following circuit ?



Q10. In the given circuit, calculate the value of current in 4.5 Ω resistor and indicate its direction. Also calculate potential difference across each cell.



Q11. In the given circuit, the voltmeter and the electric cell are ideal. Find the reading of the voltmeter (in volt)



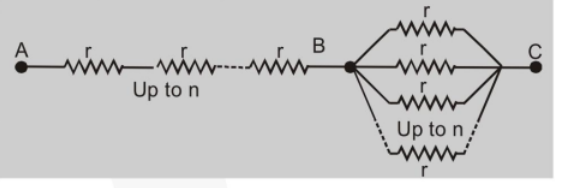
SUBJECTIVE TYPE

Q12. In hydrogen atom, the electron makes 6.62×10^{15} revolutions/sec. around the nucleus in an orbit of radius 0.5 Å. Calculate the equivalent current.

Q13. The dimension of a block is 100 m × 1 m × 1 m. The specific resistance of the material of the block is 4×10^{-6} Ω.m. Calculate its resistance across its square faces.

Q14. A battery of 10 cells each of e.m.f. E = 1.5 V and internal resistance 0.5 Ω has 1 cell wrongly connected. It is being charged by 220 V power supply with an external resistance of 47Ω in series. The potential difference across the battery.

Q15. In the following figure what is the resultant resistance between A and C ?



Q16. A galvanometer of resistance 100Ω gives full scale deflection for a current of 10^{-5} A. Calculate the shunt required to convert it into an ammeter of 1 ampere range.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The current density across a cylindrical conductor of radius R varies in magnitude according to the equation $J = J_0\left(1-\frac{r}{R}\right)$ where r is the distance from the central axis. Thus, the current denisty is a maximum J_0 at that axis (r = 0) and decreases linearly to zero at the surface (r = R). The current in terms of J_0 and conductor's cross-sectional area A is :-

(A) $\frac{J_0A}{3}$ (B) $\frac{J_0A}{6}$ (C) $\frac{J_0A}{2}$ (D) $\frac{J_0A}{5}$

Q2. A wire of cross-section area A, length L_1, resistivity ρ_1 and temperature coefficient of resistivity α_1 is connected to a second wire of length L_2, resistivity ρ_2, temperature coefficient of resistivity α_2 and the same area A, so that wire carries same current. Total resistance R is independent of temperature for small temperature change if (Thermal expansion effect is negligible).

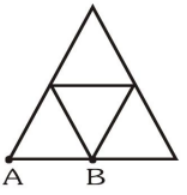
(A) $\alpha_1 = -\alpha_2$

(B) $\rho_1L_1\alpha_1 + \rho_2L_2\alpha_2 = 0$

(C) $L_1\alpha_1 + L_2\alpha_2 = 0$

(D) None

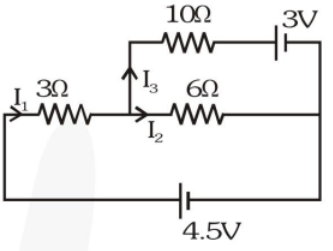
Q3. In the diagram resistance between any two junctions is R. Equivalent resistance across terminals A and B is



(A) $\frac{11R}{7}$ (B) $\frac{18R}{11}$

(C) $\frac{7R}{11}$ (D) $\frac{11R}{18}$

Q4. Find the current I_1, I_2, & I_3 in the circuit.



(A) $\frac{1}{2},\frac{1}{2},1$ (B) 1, 1, 2

(C) $\frac{1}{2},\frac{1}{2},3$ (D) $\frac{1}{2},\frac{1}{2},0$

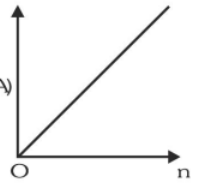
Q5. A Wheatstone's bridge is balanced with a resistance of 625 Ω in the third arm, where P, Q and S are in the 1st, 2nd and 4th arm respectively. If P and Q are interchanged, the resistance in the third arm has to be increased by 51Ω to secure balance. The unknown resistance in the fourth arm is

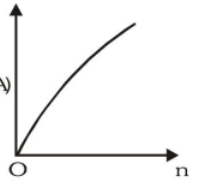


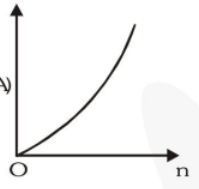
(A) 625 Ω (B) 650 Ω

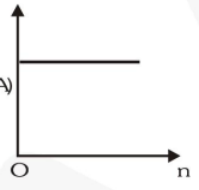
(C) 676 Ω (D) 600 Ω

Q6. A battery consists of a variable number n of identical cells having internal resistance connected in series. The terminals of the battery are short circuited and the current I measured. Which one of the graph below shows the correct relationship between I and n?

(A) 

(B) 

(C) 

(D) 

Q7. The charge flowing through a resistance R varies with time as $Q=2t-8t^2$. The total heat produced in the resistance is (for $0 \le t \le \frac{1}{8}$)

(A) $\frac{R}{6}$ joules (B) $\frac{R}{3}$ joules

(C) $\frac{R}{2}$ joules (D) R joules

Q8. Power generated across a uniform wire connected across a supply is H. If the wire is cut into n equal parts and all the parts are connected in parallel across the same supply, the total power generated in the wire is

(A) $\frac{H}{n^2}$ (B) n^2H

(C) nH (D) $\frac{H}{n}$

Q9. Two bulbs rated (25 W – 220V) and (100W – 220V) are connected in series to a 440 V line. Which one is likely to fuse?

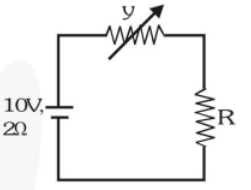
(A) 25 W bulb

(B) 100 W bulb

(C) both bulbs

(D) none

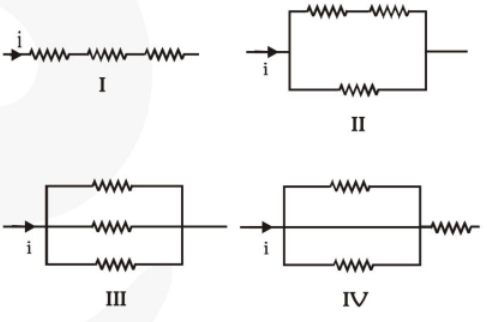
Q10. In the figure shown the power generated in y is maximum when y = 5Ω then R is -



(A) 2Ω (B) 6Ω

(C) 5Ω (D) 3Ω

Q11. The three resistance of equal value are arranged in the different combinations shown below. Arrange them in increasing order of power dissipation :



(A) III < II < IV < I

(B) II < III < IV < I

(C) I < IV < III < II

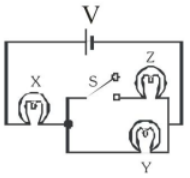
(D) I < III < II < IV

Q12. Two bulbs one of 200 volts, 60 watts & the other of 200 volts, 100 watts are connected in series to a 200 volt supply. The power consumed will be

(A) 37.5 watt (B) 160 watt

(C) 62.5 watt (D) 110 watt

Q13. If X, Y and Z in figure are identical lamps, which of the following changes to the brightness of the lamps occur when switch S is closed?



(A) X stays the same, Y decreases

(B) X increases, Y decreases

(C) X increases, Y stays the same

(D) X decreases, Y increases

Q14. A voltmeter of 998 ohm resistance is connected to a cell of emf 2 volt, having internal resistance of 2 ohms, The error in measuring emf will be :–

(A) 4×10^{-1} V (B) 2×10^{-3} V

(C) 4×10^{-3} V (D) 2×10^{-1} V

Q15. An ammeter and a voltmeter are joined in series to a cell. Their readings are A and V respectively. If a resistance is now joined in parallel with the voltmeter :-

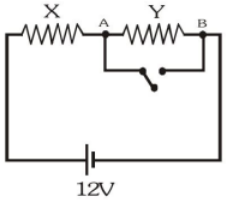
(A) both A and V will decrease.

(B) both A and V will increase.

(C) A will increase, V will decrease.

(D) A will decrease, V will increase.

Q16. When an ammeter of negligible internal resistance is inserted in series with circuit it reads 1A. When the voltmeter of very large resistance is connected across X it reads 1V. When the point A and B are shorted by a conducting wire, the voltmeters measures 10V across the battery. The internal resistance of the battery is equal to



(A) zero (B) 0.5Ω (C) 0.2Ω (D) 0.1Ω

Q17. A galvanometer has a resistance of 20Ω and reads full-scale when 0.2 V is applied across it. To convert it into a 10 A ammeter, the galvanometer coil should have a

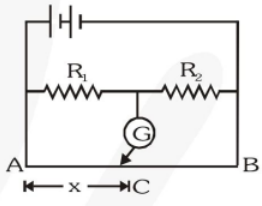
(A) 0.01Ω resistor connected across it

(B) 0.02Ω resistor connected across it

(C) 200Ω resistor connected in series with it

(D) 2000 Ω resistor connected in series with it

Q18. In the shown arrangement of the experiment of the meter bridge if AC corresponding to null deflection of galvanometer is x, what would be its value if the radius of the wire AB is doubled?

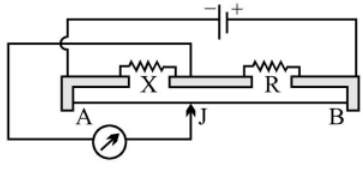


(A) x (B) x/4 (C) 4x (D) 2x

Q19. A resistance of 2Ω is connected across one gap of a metre–bridge (the length of the wire is 100 cm) and an unknown resistance, greater than 2Ω, is connected across the other gap. When these resistances are inter-changed, the balance point shifts by 20 cm. Neglecting any corrections, the unknown resistance is :-

(A) 3Ω (B) 4Ω (C) 5Ω (D) 6Ω

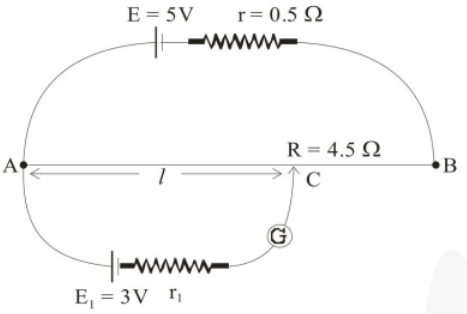
Q20. The figure shows a metre-bridge circuit, with AB = 100 cm, X = 12Ω and R = 18Ω, and the jockey J in the position of balance. If R is now made 8Ω, through what distance will J have to be moved to obtain balance?



(A) 10 cm (B) 20 cm

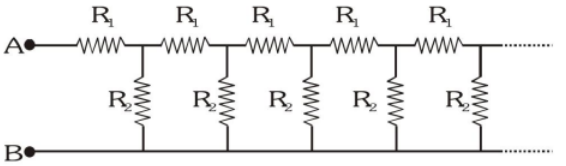
(C) 30 cm (D) 40 cm

Q21. In the given potentiometer circuit length of the wire AB is 3 m and resistance is R = 4.5 Ω. The length AC for no deflection in galvanometer is-



(A) 2 m (B) 1.8 m
(C) dependent on r_1 (D) none of these

Q22. Consider an infinite ladder network shown in figure. A voltage V is applied between the points A and B. This applied value of voltage is halved after each section. Then-



(A) $\frac{R_1}{R_2}=1$ (B) $\frac{R_1}{R_2}=\frac{1}{2}$

(C) $\frac{R_1}{R_2}=2$ (D) $\frac{R_1}{R_2}=3$

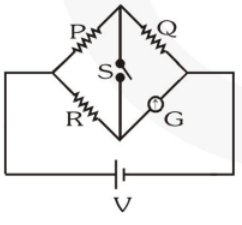
Exercise - 2A Brain Booster

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

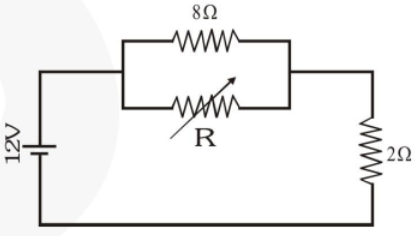
Q1. In the circuit shown P ≠ R, the reading of galvanometer is same with switch S open or closed. Then-

(A) $I_R = I_G$
(B) $I_P = I_G$
(C) $I_Q = I_G$
(D) $I_Q = I_R$



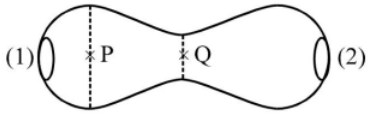
Q2. If the length of the filament of a heater is reduced by 10%, the power of the heater will
(A) increase by about 9%
(B) increase by about 11%
(C) increase by about 19%
(D) decrease by about 10%

Q3. The value of the resistance R in figure is adjusted such that power dissipated in the 2Ω resistor is maximum. Under this condition-



(A) R = 0
(B) R = 8Ω
(C) power dissipated in the 2Ω resistors is 72W
(D) power dissipated in the 2Ω resistor is 8W

Q4. A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then :



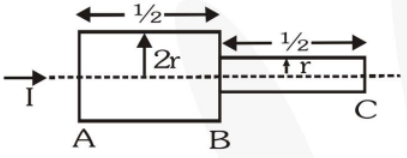
(A) the current at the cross-section P equals the current at the cross-section Q

(B) the electric field intensity at P is less than that at Q.

(C) the rate of heat generated per unit time at Q is greater than that at P

(D) the number of electrons crossing per unit area of cross-section at P is less than that at Q.

Q5. Two bars of radius r and 2r are kept in contact as shown. An electric current I is passed through the bars. Which one of following is correct :-



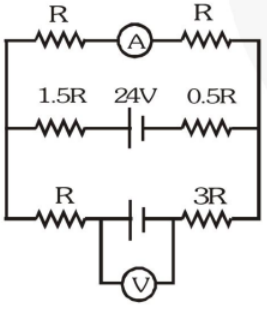
(A) Heat produced in bar BC is 4 times the heat produced in bar AB

(B) Electric field in both halves is equal

(C) Current density across AB is double that of across BC

(D) Potential difference across AB is 4 times that of across BC

Q6. If the reading of ammeter is 2A then the reading of voltmeter.



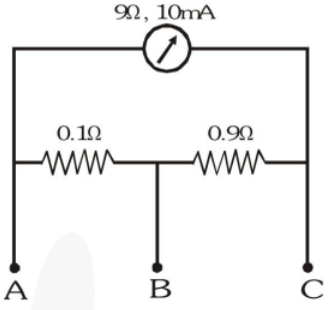
(A) Depends on R

(B) Independent on R

(C) Zero for certain value of R

(D) can't be determined

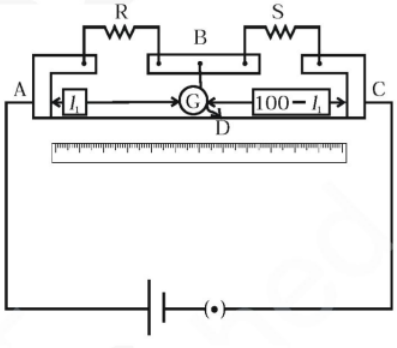
Q7. A milliammeter of range 10mA and resistance 9Ω is joined in a circuit as shown. The metre gives full–scale deflection for current I when A and B are used as its terminals, i.e., current enters at A and leaves at B (C is left isolated). The value of I is



(A) 100 mA (B) 900 mA

(C) 1A (D) 1.1 A

Q8. In a meter bridge the point D is a neutral point.



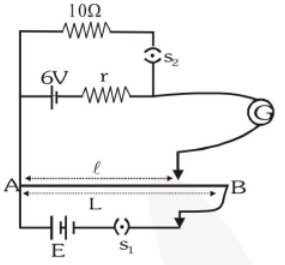
(A) The meter bridge can have no other neutral point for this set of resistances.

(B) When the jockey contacts a point on meter wire left of D, current flows to B from the wire through galvanometer.

(C) When the jockey contacts a point on the meter wire to the right of D, current flows from B to the wire through galvanometer.

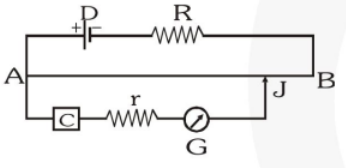
(D) When R is increased, the neutral point shifts to left.

Q9. In the arrangement shown in figure when the switch S_2 is open, the galvanometer shows no deflection for $\ell = \frac{L}{2}$. When the switch S_2 is closed, the galvanometer shows no deflection for $\ell = \frac{5L}{12}$. The internal resistance (r) of 6V cell, and the emf E of the other battery are respectively.



(A) 3Ω, 8V (B) 2Ω, 12V
(C) 2Ω, 24V (D) 3Ω, 12V

Q10. In the given potentiometer circuit, the resistance of the potentiometer wire AB is R_0. C is a cell of internal resistance r. The galvanometer G does not give zero deflection for any position of the jockey J. Which of the following cannot be a reason for this?



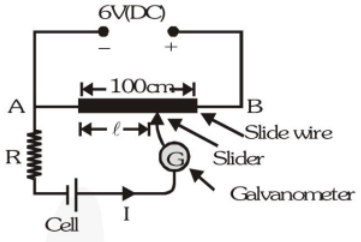
(A) $r > R_0$
(B) $R >> R_0$
(C) Emf of C> emf of D
(D) The negative terminal of C is connected to A

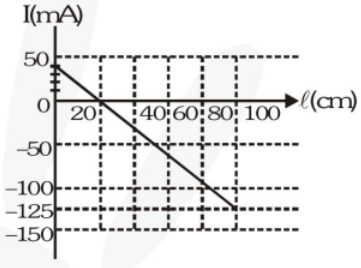
PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 11 to 13

In the circuit shown, the internal resistance of the cell is negligible. The distance of the jockey (slider) from left hand end of the wire is ℓ. The adjoining graph shows the variation of current I (marked in figure) with length ℓ of the slide wire.





Q11. For balancing condition of the instrument value of ℓ is equal to-
(A) 40 cm (B) 20 cm
(C) 100 cm (D) None of these

Q12. Value of the emf of cell, is-
(A) 0.98 V (B) 1.20 V
(C) 1.86 V (D) 3 V

Q13. Value of the resistance R, is-
(A) 30 Ω (B) 40 Ω
(C) 38 Ω (D) 45 Ω

Paragraph for Question No. 14 to 16

Important aspect of fuse wire and battery

Electric fuse a protective device used in series with an electric circuit or an electric appliance to save it from damage due to overheating produced by strong current in the circuit or appliance. Fuse wire

is generally made from an alloy of lead and tin which has high resistance and low melting point. It is connected in series in an electric installation. If a circuit gets accidentally short–circuited, a large current flows, then fuse wire melts away which causes a break in the circuit.

The power through fuse (P) is equal to heat energy lost per unit area per unit time (h) (neglecting heat loses from ends of the wire).

$$P = I^2R = h \times 2\pi r\ell \ .....\left[R = \frac{\rho\ell}{\pi r^2}\right]$$

r and ℓ are the length and radius of fuse wire. A battery is described by it's Emf (E) and internal resistance (r) Efficiency of battery is defined as the ratio of the output power and the input power

$$\eta = \frac{\text{output power}}{\text{input power}} \times 100\,\%$$

but $I = \frac{E}{R+r}$,

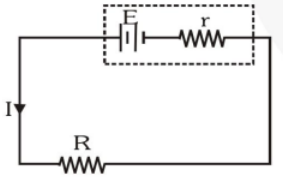
input power = EI ,

output power = $EI - I^2r$,

then $$\eta = \left(\frac{EI - I^2r}{EI}\right) \times 100 = \left(1 - \frac{Ir}{E}\right) \times 100$$

$$= 1 - \left(\frac{E}{R+r}\right)\left(\frac{r}{E}\right) \times 100 = \left(\frac{R}{R+r}\right) \times 100$$

We know that output power of a source is maximum when the external resistance is equal to internal resistance, i.e., R = r.



14. Two fuse wire of same material are having length ratio 1 : 2 and radius ratio 4 : 1. Then respective ratio of their current rating will be –

(A) 8 : 1 (B) 2 : 1

(C) 1 : 8 (D) 4 : 1

15. The maximum power rating of a 20.0 Ω fuse wire is 2.0 kW, then this fuse wire can be connected safely to a D.C. source (negligible internal resistance) of–

(A) 300 volt (B) 190 volt

(C) 250 volt (D) 220 volt

16. Efficiency of a battery (non–ideal) when delivering maximum power is–

(A) 100 % (B) 50 %

(C) 90 % (D) 40 %

Paragraph for Question No. 17 to 21

Ram and Shyam purchased two electric tea kettles 'A' and 'B' of same size, same thickness and same volume of 0.4 litre. They studied the specification of kettles as under.

Kettle – A	**Kettle – B**
Specific heat capacity = 1680 J/kg–K	Specific heat capacity = 2520 J/kg–K
mass = 200 gm	mass = 400 gm
cost = 400 Rs.	cost = 400 Rs.

When kettle 'A' is switched on with constant source, the tea begins to boil in 6 minute. When kettle 'B' is switched on with the same source separately then tea begins to boil in 8 minute.

The efficiency of kettle is defined as

$$= \frac{\text{energy used for liquid heating}}{\text{total energy supplied}}.$$

They made discussion on specification and efficiency of kettles and subsequently prepared a list of questions to draw the conclusions. Some of them are as under – (Assume specific heat of tea liquid as 4200 J/Kg–K and density as 1000 kg/m^3)

17. Efficiency of kettle 'A' is–

(A) 63.34 % (B) 83.33 %

(C) 93.34 % (D) 73.34 %

18. Efficiency of kettle 'B' is–

(A) 82.5 % (B) 72.5 %

(C) 92.5 % (D) 62.5 %

19. Ratio of electricity consumed charges for one time boiling of tea in kettle A and kettle B.

(A) 3 : 5 (B) 2 : 3

(C) 3 : 4 (D) 1 : 1

20. The resistance of coil of kettles A and B are 'R_A' and 'R_B' respectively, then we can say-

(A) $R_A > R_B$

(B) $R_A = R_B$

(C) $R_A < R_B$

(D) can not be ascertained by above data.

21. If both the kettles are joined with the same source in series one after the other. Then boiling starts in kettle 'A' and kettle 'B' after-

(A) 4 times of their original time

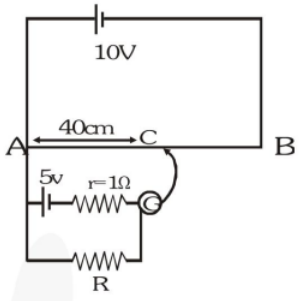
(B) equal to their original time

(C) 2 times of their original time

(D) can not be ascertained by above data

SUBJECTIVE TYPE

Q22. A potentiometer wire AB is 100 cm long and has a total resistance of 10 ohm. if the galvanometer shows zero deflection at the position C, then find the value of unknown resistance R.



Q23. Draw the circuit for experimental verification of Ohm's law using a source of variable DC voltage, a main resistance of 100 Ω, two galvanometers and two resistances of values 10^6 Ω and 10^{-3} Ω respectively. Clearly show the positions of the voltmeter and the ammeter.

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	B	C	C	C	D	B	D	A	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	B	B	D	B	B	D	C	B	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	C	A	A	A	C	B	A	B	C
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	D	D	A	D	B	B	B	B	B
41									
C									

Ex-1A

1. A **2.** A,C **3.** A,C **4.** A,C **5.** B,D

6. A,B **7.** A → (Q) ; B → (P); C → (P) **8.** 2 V; 0.4W

9. 3.7 A **10.** I = 0.5A, anticlockwise E_1= 4.25V, E_2 = 7.5V

11. 1V **12.** 10^{-3} A **13.** R = $4 \times 10^{-4}\ \Omega$ **14.** 32 V

15. $R_{AC} = \left(\frac{n^2+1}{n}\right) r$ **16.** $R_{sh} \approx 10^{-3}\ \Omega$

Ex-2

1	2	3	4	5	6	7	8	9	10
A	B	D	D	B	D	A	B	A	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	A	B	C	C	C	B	A	A	B
21	**22**								
A	B								

Ex-2A

1. A **2.** B **3.** A,C **4.** A,B,C,D **5.** A

6. A,C **7.** C **8.** A,B,C **9.** B **10.** A

11. B **12.** B **13.** A **14.** A **15.** B

16. B **17.** B **18.** D **19.** C **20.** B

21. A **22.** 4 W

-: Important Notes & Formulas :-